

## Circuit Simulation Lab: 2

### Binary Adders

#### Introduction

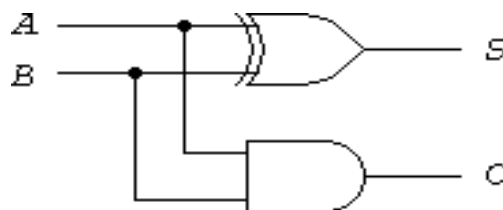
The purpose of this experiment is to introduce the design of simple combinational circuits, in this case half adders, and full adders.

#### Software tools Requirement

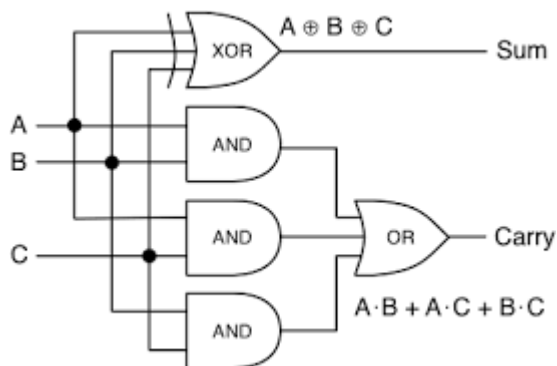
Modelsim (Siemens)

Xilinx Vivado

Logic Diagram



#### Half adder



#### Full adder

*Describe the following basic adders in Verilog HDL and capture the Waveforms*

*Dataflow modeling*

*Behavior modeling*

*Structural modeling*

*Questions to answered after this lab*

1. What is meant by combinational circuits?
2. Write the sum and carry expression for half and full adder.
3. What is signal? How it is declared?